Algorithms

Lecture # 26

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Greedy Algorithms: Activity Selection

Greedy Algorithms: Activity Selection

Here is a simple greedy algorithm that works:

- Sort the activities by their finish times.
- Select the activity that finishes first and schedule it.
- Then, among all activities that do not interfere with this first job, schedule the one that finishes first, and so on.

Greedy Algorithms: Activity Selection

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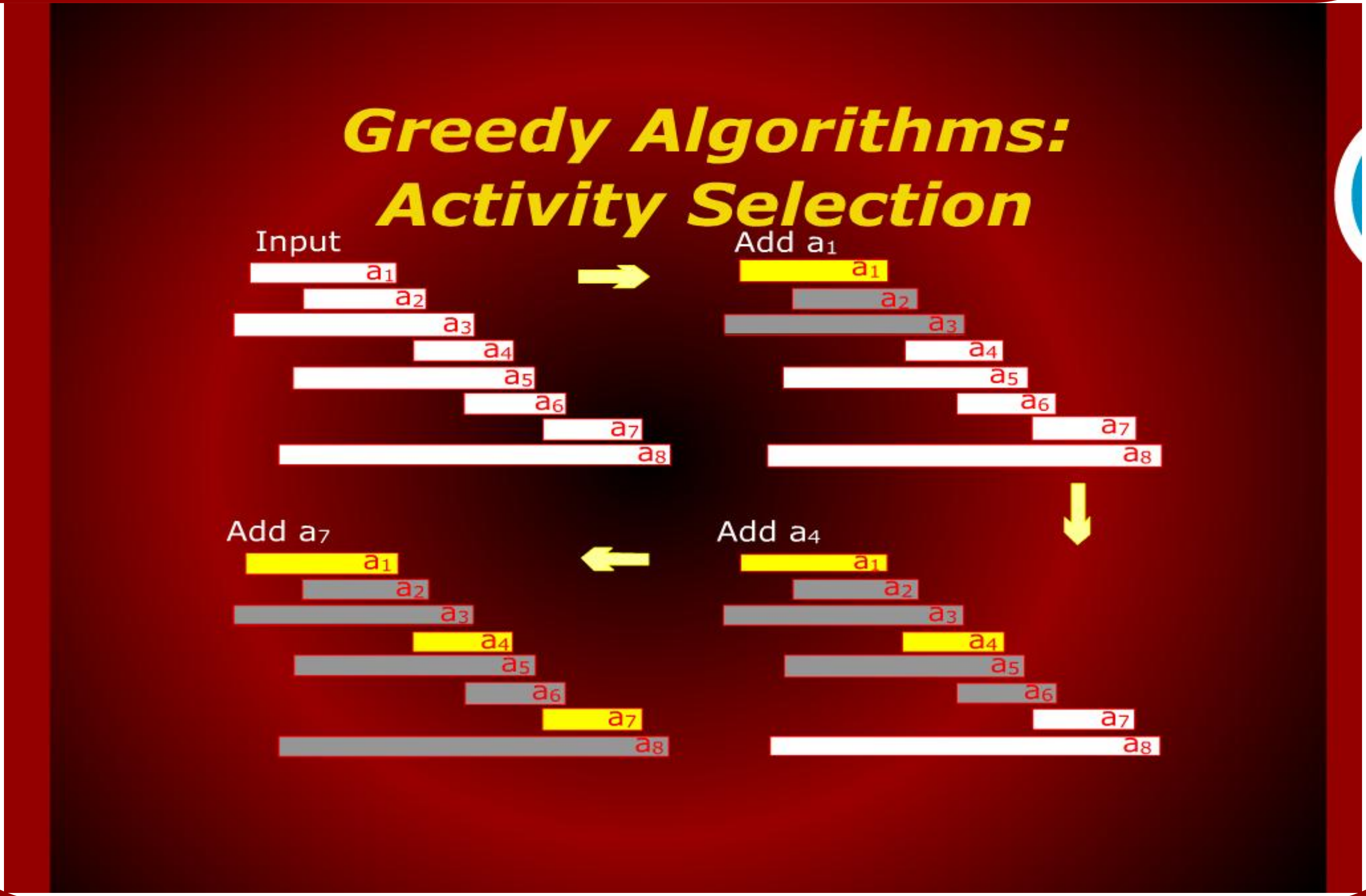
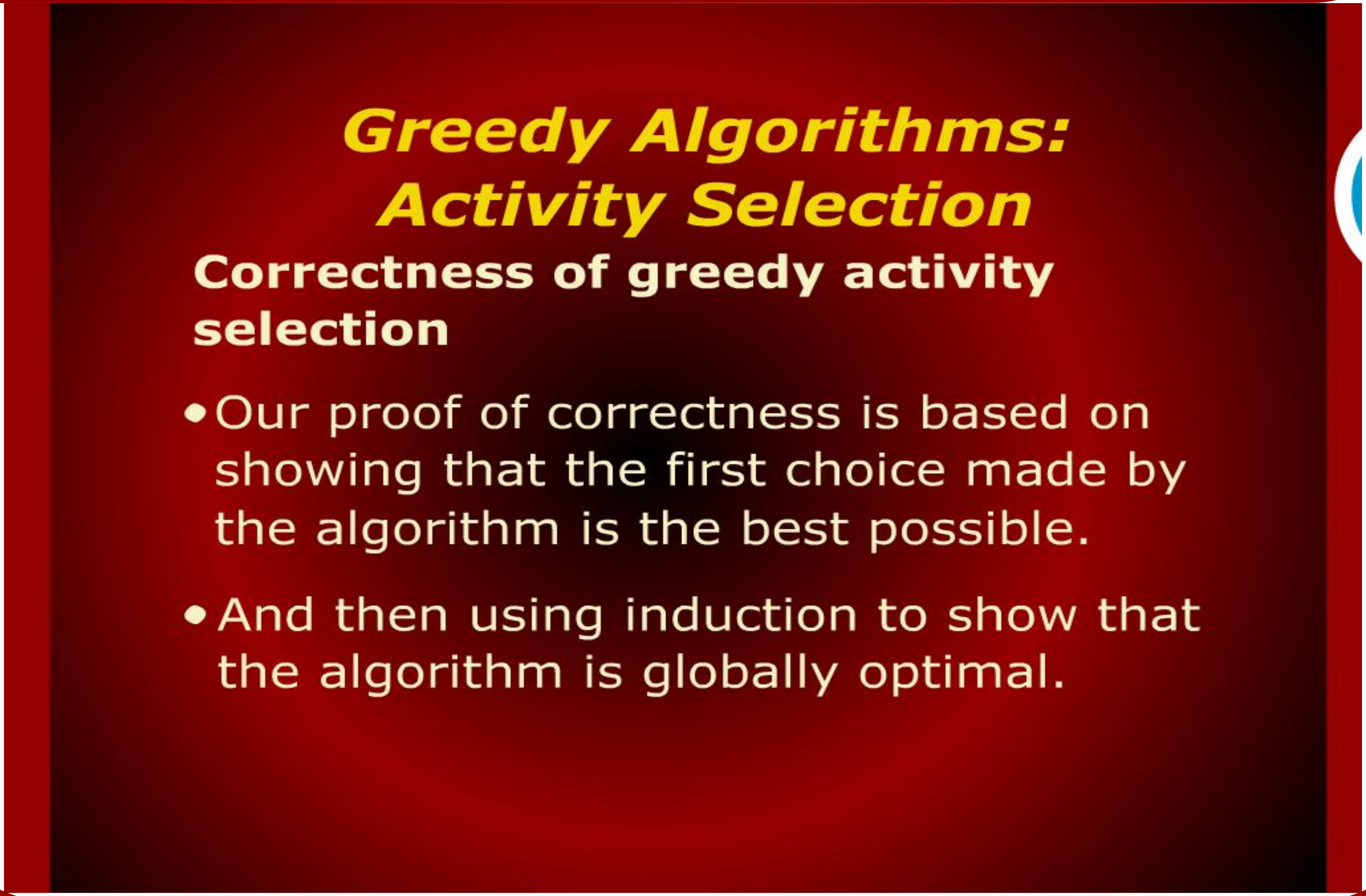
```
SCHEDULE(a[1..N])
sort a[1..N] by finish times
A ← {a[1]}; // schedule activity 1 first
prev ← 1; // most recently scheduled
for i = 2 to N
do if (a[i].start ≥ a[prev].finish)
        then A ← A ∪ a[i]; prev ← i
```

Greedy Algorithms: Activity Selection



Greedy Algorithms: Activity Selection



Greedy Activity Selection Correctness

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Correctness of greedy activity selection

- The proof structure is noteworthy because many greedy correctness proofs are based on the same idea.
- Show that any other solution can be converted into the greedy solution without increasing the cost.

Greedy Activity Selection Correctness

Greedy Activity Selection Correctness

Claim:

- Let $S = \{a_1, a_2, . . . , a_n\}$ of n activities, sorted by increasing finish times, that are to be scheduled to use some resource.
- Then there is an optimal schedule in which activity a_1 is scheduled first.

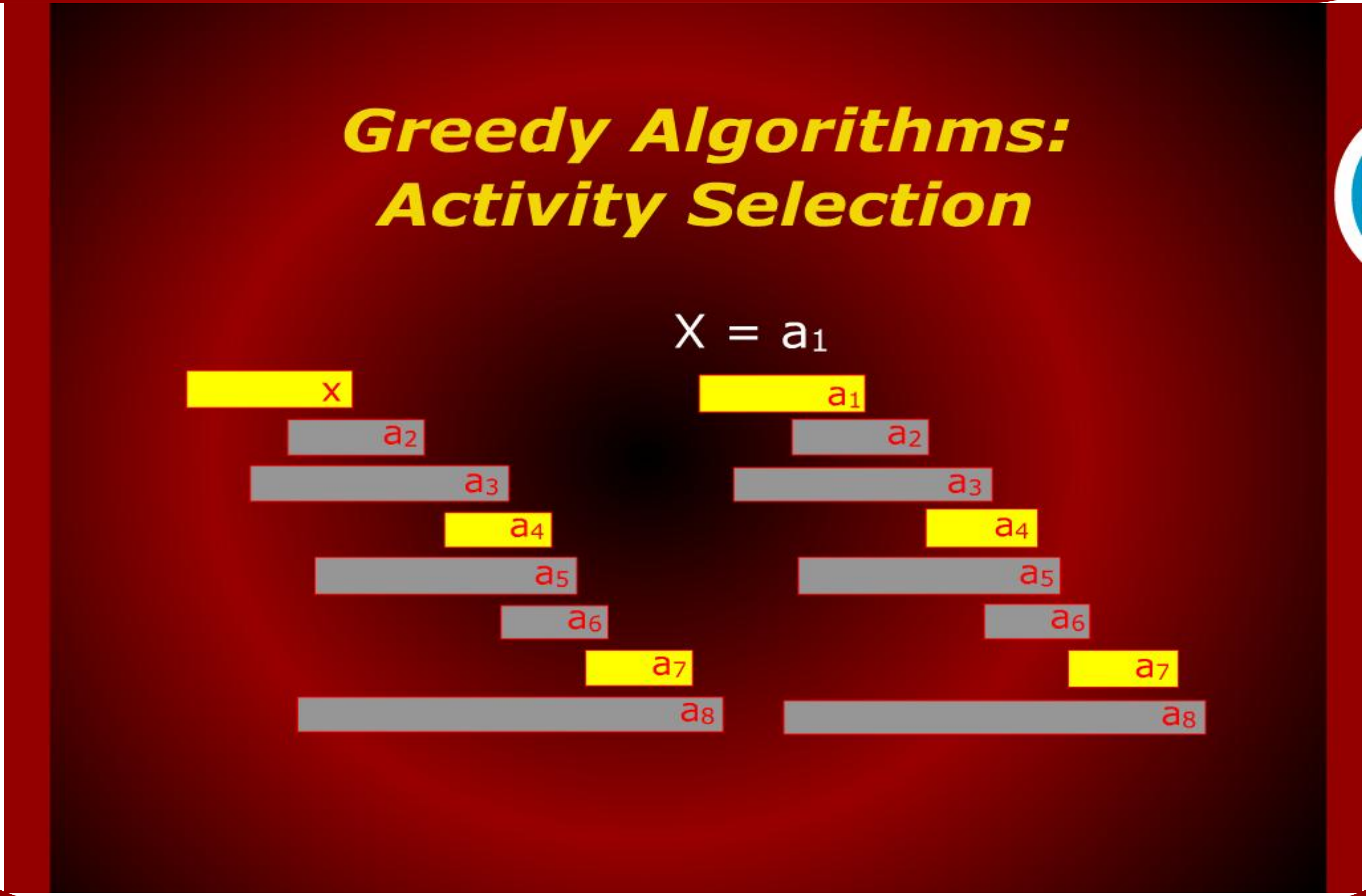
Greedy Activity Selection Correctness

Greedy Activity Selection Correctness

Proof:

- Let A be an optimal schedule.
- Let x be the activity in A with the smallest finish time.
- If $x = a_1$ then we are done.
- Otherwise, we form a new schedule A' by replacing x with activity a_1.

Greedy Algorithms: Activity Selection



Greedy Activity Selection Correctness

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Proof:

- We claim that $A' = A - \{x\} \cup \{a_1\}$ is a feasible schedule, i.e., it has no interfering activities.
- This because $A - \{x\}$ cannot have any other activities that start before x finishes, since otherwise, these activities will interfere with x.

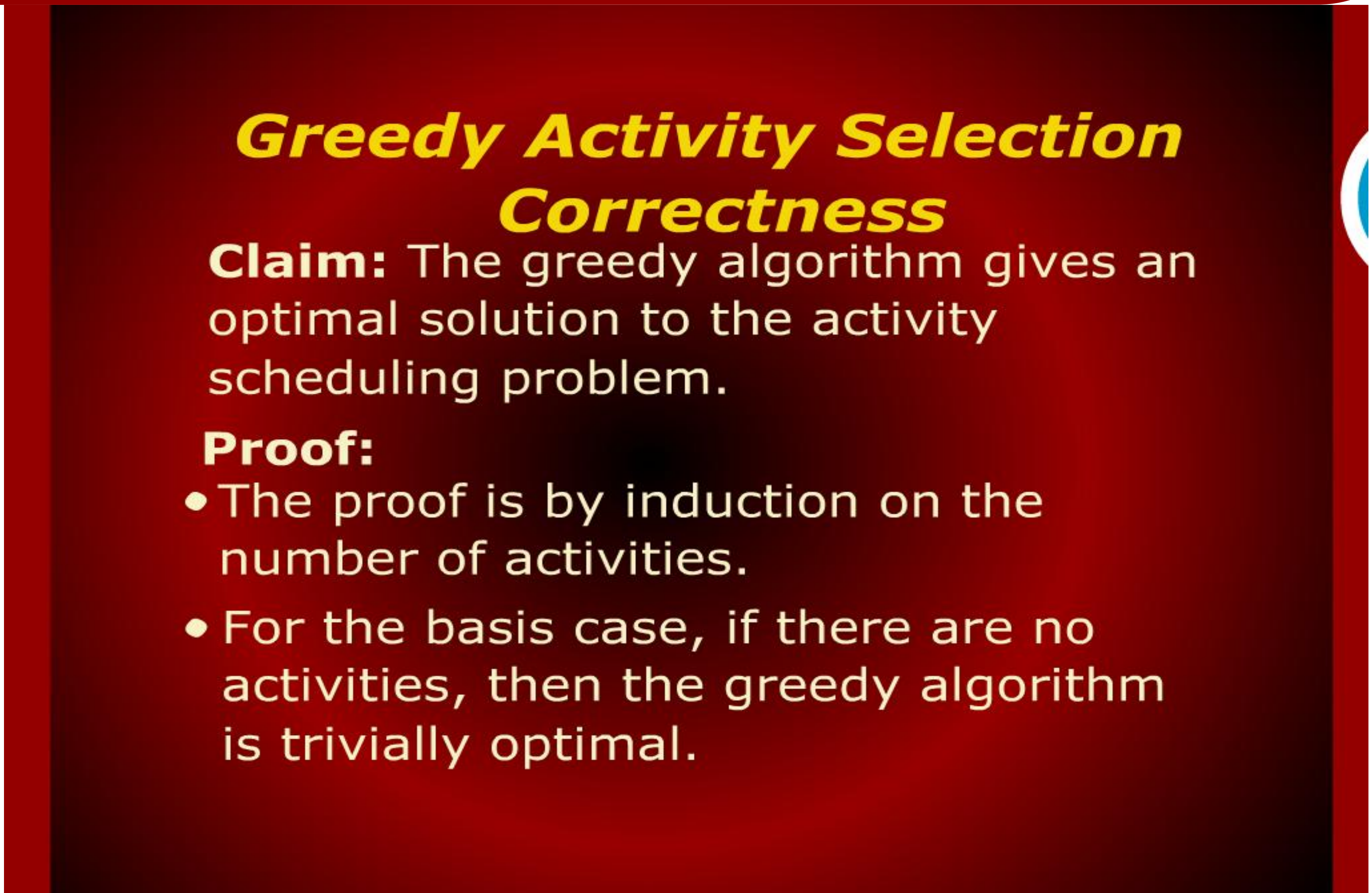
Greedy Activity Selection Correctness

Greedy Activity Selection Correctness

Proof:

- Since a_1 is by definition the first activity to finish, it has an earlier finish time than x.
- Thus a_1 cannot interfere with any of the activities in A – {x}.
- Thus, A' is a feasible schedule.
- Clearly A and A' contain the same number of activities implying that A' is also optimal.

Greedy Activity Selection Correctness



Greedy Activity Selection Optimal

Greedy Activity Selection Optimal

Proof:

- For the induction step, let us assume that the greedy algorithm is optimal on any set of activities of size strictly smaller than $|S|$ and we prove the result for S.
- Let S' be the set of activities that do not interfere with activity a_1, That is

$$S' = \{a_i \in S \mid s_i \geq f_1\}$$

Greedy Activity Selection Optimal

Greedy Activity Selection Optimal

Proof:

- Any solution for S' can be made into a solution for S by simply adding activity a_1, and vice versa.
- Activity a_1 is in the optimal schedule (by the above previous claim).

Greedy Activity Selection Optimal

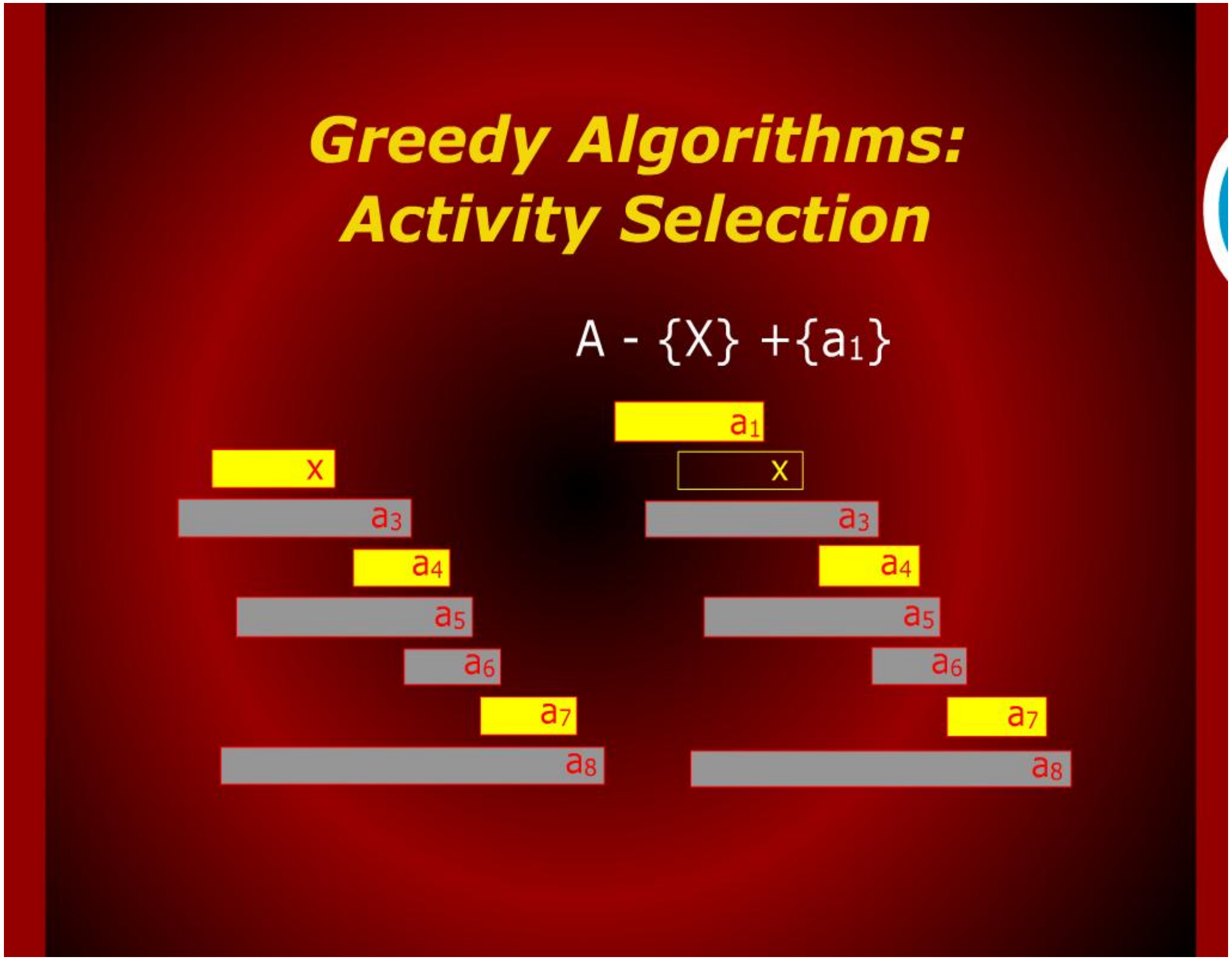
Greedy Activity Selection Optimal

Proof:

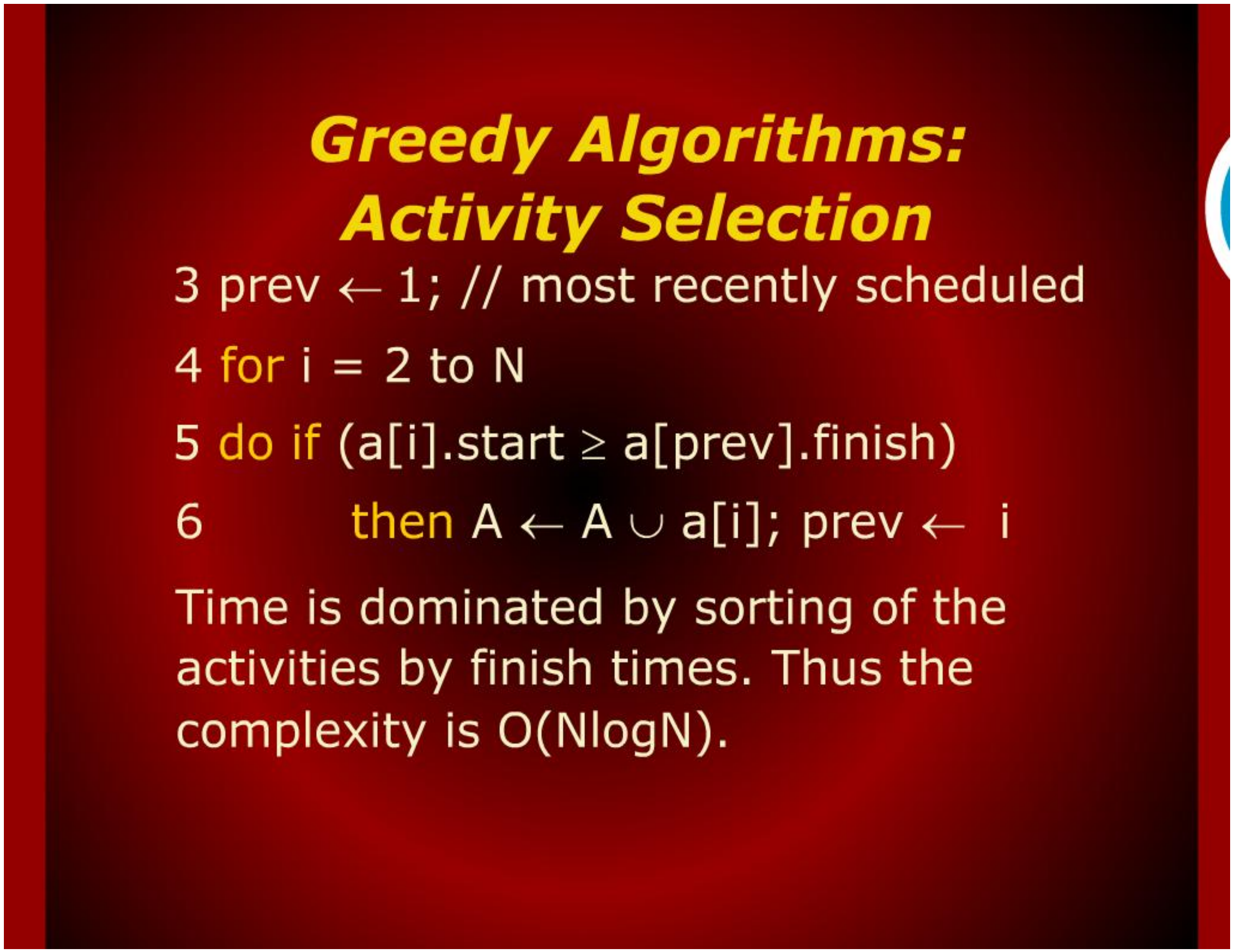
- It follows that to produce an optimal schedule for the overall problem, we should first schedule a_1 and then append the optimal schedule for S'.
- But by induction (since $|S'| < |S|$), this exactly what the greedy algorithm does.

Activity Selection

Greedy Algorithms: Activity Selection



Greedy Algorithms: Activity Selection



Random Access Machine